

DAY 10

NAME: Nandhini V

DATE: 18-08-2025

Task:

- Create Fast API, to store, retrieve all and retrieve specific song by Id.
- SQLite can be used as data store
- table can contain
 - song name, playing time in seconds, album (or movie) along with Id

Code:

songs.html

```
<html>
  <head>
    <title>Songs List</title>
    <style>
      table {
        width: 60%;
        border-collapse: collapse;
        margin: 20px auto;
      }
      th, td {
        padding: 10px;
        border: 1px solid black;
        text-align: center;
      }
      th {
        background: #8c98ce;
      }
    </style>
  </head>
  <body>
    <h2 style="text-align:center;">Songs List</h2>
    <table>
      <tr>
        <th>Id</th>
        <th>Name</th>
        <th>Movie</th>
        <th>Duration</th>
      </tr>
```

```

        {% for song in songs %}
        <tr>
            <td>{{ song.Id }}</td>
            <td>{{ song.Name }}</td>
            <td>{{ song.Movie }}</td>
            <td>{{ song.Duration }}</td>
        </tr>
        {% endfor %}
    </table>
</body>
</html>

```

Day10Task.py

```

from sqlalchemy import create_engine
from sqlalchemy.dialects.sqlite import *
SQLALCHEMY_DATABASE_URL = "sqlite:///./song_db.db"

```

```

engine = create_engine(SQLALCHEMY_DATABASE_URL, connect_args =
{"check_same_thread" : False})

```

```

from sqlalchemy.orm import sessionmaker, Session, declarative_base
session = sessionmaker(autocommit = False, autoflush = False, bind = engine)

```

```

Base = declarative_base()

```

```

from sqlalchemy import Column, Integer, String

```

```

class songs(Base):
    __tablename__ = 'songs'
    Id = Column(Integer, primary_key=True, index=True)
    Name = Column(String(50), nullable=False)
    Movie = Column(String(30), nullable=False)
    Duration = Column(Integer, nullable=False)
    Base.metadata.create_all(bind=engine)

```

```

from fastapi import FastAPI, Request, Depends
import uvicorn
from fastapi.templating import Jinja2Templates
app = FastAPI()

```

```

templates = Jinja2Templates(directory="my_web_pages")

```

```

from pydantic import BaseModel

```

```

class song(BaseModel):
    Id:int
    Name: str
    Movie:str

```

```

Duration: int
class Config:
    orm_mode = True

def get_db():
    db = session()
    try:
        yield db
    finally:
        db.close()

@app.post('/add_song')
def add_song(s: Song, db: Session = Depends(get_db)):
    sg = Song(Name=s.Name, Movie=s.Movie, Duration=s.Duration)
    db.add(sg)
    db.commit()
    db.refresh(sg)
    return sg

@app.get("/songs_list")
def view_songs(request: Request, db: Session = Depends(get_db)):
    records = db.query(Song).all()
    return templates.TemplateResponse("songs.html", {"request": request, "songs": records})

@app.get("/getBy/{_id}")
async def getBy(_id: int, db: Session = Depends(get_db)):
    sg = db.query(Song).filter_by(id = _id).first()
    return sg

if __name__ == "__main__":
    uvicorn.run("Day10Task:app", host = "127.1.1.0", port = 8000, reload = True)

```

Output:

Before Adding - Database

Table: songs				
	<u>Id</u>	Name	Movie	Duration
	Filter	Filter	Filter	Filter
1	1	Hosanna	VTV	300
2	2	Kanimaa	Retro	400
3	3	Sawade...	Vidaamuyar...	250
4	4	Mundhi...	Vaaranam ...	300
5	5	Kannai ...	Iru Mugan	400

Request body required

application/json

Edit Value | Schema

```
{
  "Id": 0,
  "Name": "Jinguchaa",
  "Movie": "Thug Life",
  "Duration": 350
}
```

Execute

Clear

Server response

Code

Details

200

Response body

```
{
  "Name": "Jinguchaa",
  "Movie": "Thug Life",
  "Duration": 350,
  "Id": 6
}
```

Download

Response headers

```
content-length: 62
content-type: application/json
date: Mon, 18 Aug 2025 12:34:39 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header

Example Value | Schema

```
"string"
```

List Of Songs - UI

Songs List

Id	Name	Movie	Duration
1	Hosanna	VTV	300
2	Kanimaa	Retro	400
3	Sawadeeka	Vidaamuyarchi	250
4	Mundhinam	Vaaranam Aayiram	300
5	Kannai Vittu	Iru Mugan	400
6	Jinguchaa	Thug Life	350

After adding - Database

Table: songs

	<u>Id</u>	Name	Movie	Duration
	Filter	Filter	Filter	Filter
1	1	Hosanna	VTV	300
2	2	Kanimaa	Retro	400
3	3	Sawade...	Vidaamuyar...	250
4	4	Mundhi...	Vaaranam ...	300
5	5	Kannai ...	Iru Mugan	400
6	6	Jinguc...	Thug Life	350

Retrieve specific song by Id

Name	Description
_id * required integer (path)	3
Execute Clear	

Server response

Code	Details
200	Response body <pre>{ "Name": "Sawadeeka", "Movie": "Vidaamuyarchi", "Duration": 250, "Id": 3 }</pre> Response headers <pre>content-length: 66 content-type: application/json date: Mon, 18 Aug 2025 12:38:31 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links

Media type:

Controls Accept header:

Example Value | Schema

```
"string"
```

← → ↺ ⓘ 127.1.1.0:8000/getBy/3

Pretty-print ☒

```
{
  "Name": "Sawadeeka",
  "Movie": "Vidaamuyarchi",
  "Duration": 250,
  "Id": 3
}
```